Strain-specific GEM reconstruction pipeline: concept

Draft v0.1 · 2026-10-22 · assemblies, AMRBench_QC gene prediction, CarveMe, then species-reference biomass and gap-filling

PER GENOME · 140 NOW, 12,000 LATER (7 SPECIES)

Assemblies

`data/genomes/*/*.fna`
samplesheet:
`sample, species, fasta`
species selects the reference

AMRBench_QC (Nextflow) D1

`gigabiome/AMRBench_QC @ 52001eb`
QUAST 5.3.0 · assembly statistics
CheckM2 1.1.0 · completeness, contamination
Bakta 1.12.0 · gene prediction + annotation
Prokka 1.15.6 · optional (--skip_prokka)
Output: proteins (.faa), GFF3, QC reports

QC gate D2
pass?

fail → **Excluded from modelling**
logged with reason

pass: proteins (.faa) + species

Orchestration (applies to every per-genome step) D8

Nextflow DSL2, one task per genome, -resume
One pinned container per tool; all versions recorded
Independent per-genome tasks: scales to 12,000 on HPC
MILP time limit / optimality gap recorded per model
Species-level preparation runs once and is reused

CarveMe 1.6.6 · carve D7 D6

DIAMOND: proteins vs CarveMe BiGG DB
Reaction scores from BiGG GPRs
MILP carving of the universe model
gramneg (Gram–) / grampos (Gram+)
--reference: score bonus for reactions shared with the species reference
no -g: gap-filling done downstream
Output: draft model (universal 'Growth')

Biomass replacement

Remove universal 'Growth' reaction
Insert the species reference biomass
Add reactions making its precursors that the draft lacks (lipids, cell wall, lumped pseudo-metabolites, e.g. iYS854)
Set as the model objective

Reference-guided gap-filling D3 D5 D9

MILP (CarveMe gapFill algorithm)
Pool = harmonised reference + universe
Cost: reference < universe reactions; homology-scored reactions cheaper
Growth ≥ minimum on the reference biomass, in the chosen medium
Each added reaction tagged with its source (reference / universe)

Model QC & provenance

Valid SBML; loads in COBRApy
Grows on the gap-fill medium
Biomass precursors producible
Mass and charge balance check
Gap-fill size, gene count
Tool versions and parameters written into the SBML notes

Outputs

`<sample>.xml`
`<sample>.json`
gap-fill report
QC table row
run summary

reference · *biomass module* · *pool, costs, medium* · *medium, biomass*

ONCE PER SPECIES (7×) · REUSED BY EVERY GENOME OF THAT SPECIES

Reference registry D4

`references.tsv`
species: reference GEM (SBML)
biomass reaction ID
CarveMe universe: gramneg / grampos
gap-filling medium
reference choice per species

Prepare reference (once per species)

Map reference IDs and compartments to CarveMe's BiGG namespace (e.g. iYS854 'w' compartment)
Extract the biomass reaction and the reactions that synthesise its precursors
Build the gap-fill pool = reference + universe reactions, with a cost per reaction
Define the medium as BiGG exchange reactions
Check: harmonised reference grows on the medium with its own biomass
Check: reference biomass is producible from the pool

Prepared reference bundle (per species)

harmonised reference model (SBML)
biomass module: biomass + precursor reactions
gap-fill pool with reaction costs
medium definition
validation report

OPEN DECISIONS · TO SETTLE BEFORE IMPLEMENTATION

- D1 Protein source for CarveMe: Bakta .faa or Prokka .faa
- D2 QC thresholds: CheckM2 completeness / contamination, QUAST metrics
- D3 Gap-filling medium: LB, M9 or species-specific; one or several
- D4 Reference per species: E. coli and A. baumannii choices; Enterobacter has none
- D5 Gap-fill costs: reference vs universe vs homology-scored reactions
- D6 MILP solver: CPLEX / Gurobi (licence) or SCIP (600 s limit)
- D7 CarveMe --reference-score (default 0.0)
- D8 Compute environment: HPC scheduler or cloud; container engine
- D9 Minimum growth rate for gap-filling (CarveMe default 0.1)

LEGEND

- Existing tool / pipeline
- Custom step (to be built)
- Data / inputs / outputs
- QC gate
- Dn Open decision
- Species reference used by a step

Verified against CarveMe 1.6.6 source: --reference changes only the carving scores of reactions shared with the reference; -g gap-fills from the universe model only; the biomass is the universe 'Growth' reaction. The orange steps implement the requested reference biomass and reference-based gap-filling, which CarveMe does not provide. Tool versions for AMRBench_QC are taken from its module definitions at commit 52001eb.